

Vignesh T

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Summary

Computer Science Engineering student specializing in network security, applied cryptography (RSA, AES-256, post-quantum Kyber), and AI systems (RAG, LangChain, FAISS). Skilled in Python, Linux, REST APIs, Docker, and CI/CD, with experience troubleshooting complex systems, automating workflows, and delivering end-to-end solutions.

Education

Acharya Institute of Technology (VTU)

Bengaluru, Karnataka

Bachelor of Engineering in Computer Science and Engineering; CGPA: 8.3/10

Aug 2023 – Jun 2027

- Coursework: Data Structures & Algorithms, Computer Networks, Operating Systems, Database Management Systems

Technical Skills

Operating Systems & Networking: Linux/Unix, TCP/IP fundamentals, Socket Programming, Network Security

Languages & Databases: Python, C, C++, Java, MySQL, SQL

Tools & Infrastructure: Git, GitHub, VS Code, Jupyter Notebook, Flask, REST APIs, Docker, CI/CD

AI / ML: Retrieval-Augmented Generation (RAG), LangChain, FAISS, Prompt Engineering

Security: AES-256, RSA, Post-Quantum Cryptography (Kyber)

Experience

Earthy

Bengaluru, Karnataka

Founder's Associate (Growth Engineer)

Oct 2025 – Oct 2026

- Architected and deployed internal business operating systems, including the company website, UI/UX workflows, and end-to-end technical infrastructure, to accelerate startup growth
- Resolved technical issues across product and operations, documenting fixes and supporting non-technical teams
- Owned full-cycle product, engineering, and operations tasks, improving team efficiency and customer engagement
- Partnered directly with founders to define strategic roadmaps and execute business development operations

Projects

Quantum-Resistant Secure Communication System | Python, AES-256, RSA, Kyber, PQC

GitHub | 2026

- Designed and implemented a client-server communication framework over TCP sockets, benchmarking classical RSA-2048 + AES-256-GCM against post-quantum Kyber-1024 + AES-256-GCM
- Built secure session establishment and real-time encrypted communication; analyzed key generation latency, encryption/decryption throughput, and computational overhead

Advanced RAG Knowledge Assistant | Python, LangChain, FAISS, Flask, Streamlit

GitHub | 2026

- Engineered a Python-based Retrieval-Augmented Generation pipeline with document ingestion, semantic chunking, transformer-based embeddings, and FAISS vector database retrieval
- Delivered an interactive Streamlit interface with real-time analytics, source attribution, and confidence estimation across custom document sources

Production E-Commerce Platform | REST APIs, Database Transactions, Docker, CI/CD

GitHub | 2026

- Built a full-stack e-commerce platform with user authentication/authorization, a product catalog with category-based search and inventory tracking, and a shopping cart and checkout flow with sandboxed payment integration
- Implemented order management and an admin dashboard backed by transactional database operations, exposed through REST APIs and validated with automated testing
- Containerized the application with Docker and configured a CI/CD pipeline for automated build, test, and deployment

Certifications

Automating Cyber Security with AI – LinkedIn Learning, 2026

RAG AI Apps and AI Agent for Cybersecurity and Networking – LinkedIn Learning, 2026

Introduction to Applied Cryptography and Cryptanalysis – LinkedIn Learning, 2026

Achievements & Extracurricular Activities

Hackathon Participant – Prototyped and delivered solutions under tight deadlines (2023 – Present)